SHAHADAT HUSSAIN, PhD

MATERIALS & ADDITIVE MANUFACTURING ENGINEER

Abu Dhabi, UAE | shahadat.hussain@aol.com | linkedin.com/in/shahadathussain | shahadathussain.com

PROFESSIONAL SUMMARY

Materials and Manufacturing Engineer with 6+ years delivering additive manufacturing (LPBF, FFF, SLA) process development, materials qualification, and mechanical testing programs across academic and applied research settings in the UAE. Hands-on operator of 10+ industrial 3D printing systems and a full metallurgical characterization and testing suite (SEM, XRD, DSC, Instron, MTS). Specialist in metal and polymer-composite lattice structures (TPMS), nickel-titanium and copper-based shape memory alloys, and process-structure-property qualification for lightweight, high-performance components. Combines materials engineering depth with Python/MATLAB-based data analysis to support process optimization and quality mapping. Seeking to apply this qualification and testing expertise to industrial R&D, manufacturing engineering, or materials qualification roles in R&D or advanced manufacturing sector.

CORE COMPETENCIES

- Additive Manufacturing Process Development (LPBF, FFF, SLA)
- Materials Characterization (SEM, XRD, DSC, Optical Microscopy)
- Mechanical Testing & Qualification (Static, Fatigue, Compression)
- NiTi & Cu-Based Shape Memory Alloys
- TPMS / Architected Lattice Structures
- Process Parameter Optimization & Defect/Quality Mapping
- Composite & Metal AM (Carbon-Fiber PLA/PPA, Nylong, ABS, Verowhite, Metal Powder Bed)
- CAD & Build Preparation (FreeCAD, Materialise Magics, GrabCAD)
- Data Analysis & Process Modeling (Python, MATLAB)
- EHS / Lab Safety & Risk Assessment

PROFESSIONAL EXPERIENCE

Research Associate — Additive Manufacturing & Advanced Materials *Sep 2024 – Present*

New York University Abu Dhabi (NYUAD), Abu Dhabi, UAE

- Delivered carbon-fiber-reinforced PLA/PPA composite metamaterial components (gyroid and primitive TPMS lattices) via FFF for aerospace, automotive, and structural application targets.
- Qualified printed composite structures against quasi-static compression, flexural, modal, and frequency-response test protocols to validate strength, energy absorption, and vibration performance.
- Operated and maintained 10+ industrial additive manufacturing systems, including Raise3D Pro3 Plus HS, Stratasys J850/F370, Formlabs Form3, and Bambu Lab Carbon X1, supporting continuous production and prototyping throughput.

Postdoctoral Research Fellow — Metal 3D Printing & Smart Materials *Feb 2020 – Jun 2024*

Khalifa University, Abu Dhabi, UAE

- Fabricated and qualified NiTi shape memory alloy TPMS lattice structures via Laser Powder Bed Fusion (EOS M400); optimized laser power, scan speed, and energy density to control microstructure and phase transformation behavior.
- Executed full materials characterization pipeline on 50+ printed samples using JEOL 7610F SEM, FEI Quanta 3D FIB-SEM (SEI/BSE/EDS), Bruker D2 and Rigaku MiniFlex XRD, Mettler Toledo and Setaram DSC, and Leica optical microscopy.
- Ran static (Instron 5969, 50 kN), fatigue (Instron 8872, 25 kN), and compression (MTS 100 kN) qualification campaigns; developed defect and print-quality maps used to reduce porosity and improve part consistency in AM NiTi components.

- Authored 4 peer-reviewed publications and supported 450+ citations of program-relevant materials data (Materials & Design, Journal of Manufacturing Processes, Materials, JMEP); reviewed 30+ technical manuscripts across 15 journals.
- Completed comprehensive EHS lab safety training covering chemical handling, HF acid etching, cryogenic substances, risk assessment, and emergency response protocols.

Project Fellow — Smart & Functional Materials *Jun 2013 – Aug 2014*

CSIR – Advanced Materials & Processes Research Institute (AMPRI), Bhopal, India

- Synthesized Cu-Al-Mn and Cu-Al-Ni shape memory alloys via vacuum induction melting under a Government of India (12th Five Year Plan) funded program; performed hot/cold rolling, heat treatment, quenching, and metallographic preparation.
- Characterized phase transformation and mechanical properties using SEM, XRD, DSC, and hardness/tensile testing; outputs contributed to 5 peer-reviewed publications and a research collaboration with IIT Madras.

Engineering Intern — Steel Plant Operations *Jun – Jul 2011*

Bokaro Steel Plant, Steel Authority of India Limited (SAIL), Jharkhand, India

- Gained direct industrial exposure to large-scale integrated steel manufacturing processes, plant operations, and quality systems.

EDUCATION

PhD, Materials Science & Technology *2014 – 2019*

Academy of Scientific and Innovative Research (AcSIR) / CSIR-AMPRI, Bhopal, India — Thesis: Cu-based shape memory alloys: composition, processing, microstructure, and shape memory properties

BEng, Mechanical Engineering *2008 – 2012*

UIT, Rajiv Gandhi Proudyogiki Vishwavidyalaya (RGPV), Bhopal, India

TECHNICAL SKILLS

Additive Manufacturing Systems: EOS M400 (LPBF/SLM); Stratasys F370/J850 (FDM/PolyJet); Raise3D Pro3 Plus HS, Forge1 (FFF); Formlabs Form3 (SLA); Ultimaker S1

Materials Characterization: JEOL 7610F SEM; FEI Nova NanoSEM; FEI Quanta 3D FIB-SEM (SEI/BSE/EDS); Bruker D2 & Rigaku MiniFlex XRD; Mettler Toledo DSC1; Setaram DSC; Leica Optical Microscopy; Bruker Q4 Tasman

Mechanical Testing & Processing: Instron 5969 (static, 50 kN); Instron 8872 (fatigue, 25 kN); MTS 100 kN UTM; Controls Simplex 350 & Automax; vacuum induction melting; two-roll rolling; heat treatment; metallography; etching

Design, Simulation & Computing: SolidWorks; FreeCAD; Materialise Magics; GrabCAD Print; Bambu Studio; PreForm; IdeaMaker; Cura; Python (Pandas, NumPy, Scikit-learn); MATLAB; SQL; Linux; Shell scripting; LaTeX

CERTIFICATIONS

- Additive Manufacturing Specialization — Arizona State University
- IBM Data Science Professional Certificate — IBM
- Responsible Conduct of Research — CITI Program, Florida
- Google AI Essentials Certificate — Google
- Python for Everybody Specialization — University of Michigan
- Lab Safety (EHS) — Khalifa University

SELECTED PUBLICATIONS

500+ citations | h-index: 10 | ORCID: 0000-0002-4355-2169 | Peer reviewer for 15 journals including Additive Manufacturing, Materials & Design, Scripta Materialia

- Hussain S, Mehraj H, Karathanasopoulos N, Moustafa MA. Do Chopped Fibers Matter? Process-Structure-Property Relationships and Thermal Annealing Effects in Fiber-Reinforced TPMS Lattices. Progress in Additive Manufacturing, 2026.

- Hussain S, Alagha AN, Zaki W. Phase Transformation Behavior of NiTi TPMS Lattices Fabricated by LPBF. Journal of Materials Engineering & Performance, 2024.
- Hussain S, Alagha AN, Haidemenopoulos GN, Zaki W. Microstructural & Surface Analysis of NiTi TPMS Lattice Sections Fabricated by LPBF. Journal of Manufacturing Processes, 2023.

PROFESSIONAL MEMBERSHIPS

- Technical Volunteer, Materials Research Society (MRS)
- Individual Member, Research Data Alliance (RDA)

ADDITIONAL INFORMATION

Languages: English (Professional), Hindi (Native), Urdu (Native)

Profiles: github.com/drshahadathussain | researchgate.net/profile/Shahadat-Hussain | scholar.google.com/citations?user=tQNSWaAAAAAJ